

InventOR: A Prompt-Configured, No-Fine-Tuning LLM Workflow for Material-Level Inventory Control

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Abstract

This paper presents *InventOR*, a prompt-configured large-language-model (LLM) workflow for material-level inventory decision support under demand and lead-time uncertainty. A material-scoped historical CSV and inline parameter block are supplied through an enterprise LLM application; no project-specific model weights were trained or fine-tuned on enterprise data. The LLM returns a policy artifact that deterministic parsing and simulation evaluate against an SAP-derived comparator and an SAP-safety-lead-time-informed operations-research (r, Q) comparator. The LLM is therefore a candidate-policy generator, not an autonomous ERP controller. The executed artifact set covers all 365 eligible plant-material pairs in a three-plant industrial dataset after retry handling and capacity validation. A common-feasible cohort of 346 pairs applies identical MOQ and per-material storage rules to all arms. Working-day post-cutoff simulation reports 77.52% demand-weighted aggregate fill, 97.94% mean material fill, and 2,848.96 mean simulated material-horizon holding/order cost for the LLM-emitted arm. Positive lost-sales penalty sensitivities remove its

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apparent cost advantage. These values remain descriptive simulated comparison rather than controlled production-trial evidence. Stored metadata do not establish the exact deployed prompt or model/runtime identity; the contribution is a transparent prototype workflow and disclosure of controls required for prospective evaluation.

Keywords: Generative Artificial Intelligence, Large Language Models, Operations Research, Supply Chain Management, Decision Support Systems, Explainable AI, Inventory Control

1. Introduction

Supply chain planning operates under volatile demand, uncertain replenishment, and disruption exposure [1]. Many material-control decisions nevertheless still depend on simple parameters: reorder points, safety stock, order quantities, review periods, and sometimes safety lead time; these settings strongly affect service and inventory exposure [2, 3]. Classical operations research offers rigorous tools for stochastic and robust inventory decisions [4, 5, 6], but operational planners often need a narrower capability: converting imperfect historical records into transparent, defensible, and auditable material-level policy values [7].

Large language models (LLMs) can help connect business intent, operational evidence, and analytical workflows [8, 9]. In inventory control, however, recommendations require explainability, reproducibility, accountability, and meaningful human control before planners can act on them [10, 11]. This creates a practical design requirement for hybrid AI-supported OR systems: LLM artifacts must be inspectable, while their numeric recommendations require explicit simulation and human release controls [9, 11].

18 This paper addresses that requirement through *InventOR*, an operations-
19 research-first workflow for material-level inventory decision support. An
20 enterprise LLM application produces policy artifacts from material-scoped
21 inputs; deterministic software parses and simulates their numeric values.

22 The contribution of this paper is an evaluation protocol rather than a
23 new inventory model or solution algorithm. The comparator policies are
24 standard textbook constructions; what the paper adds is a reusable, au-
25 ditable procedure for deciding whether LLM-emitted policy values are ad-
26 missible and how they perform against ERP-derived baselines on a common
27 simulation horizon. Three components make up that protocol.

28 **C1 Admissibility protocol for LLM-emitted policy artifacts.**

29 Material-specific inputs pass through a configured enterprise LLM ap-
30 plication without project-specific training or fine-tuning. Deterministic
31 scoreability and capacity checks decide admission; the architecture
32 reserves a planner release step, which this study documents as a design
33 requirement rather than an evaluated intervention [12].

34 **C2 Executable boundary specification.** The paper documents the
35 material-input contract, alias-aware conversion, capacity validation,
36 and simulation boundary, so that a competing generator can be sub-
37 stituted and scored under identical rules.

38 **C3 Complete-cohort reporting discipline.** All 365 eligible pairs from
39 233,078 parsed records yielded scoreable, capacity-feasible artifacts af-
40 ter retry handling; 346 pairs meet the common feasibility rules used
41 for the three-arm descriptive comparison. No pair is silently dropped,
42 and every exclusion rule is prespecified and reported.

43 The remainder of the paper reviews related work, presents the executed
44 workflow and policy logic, reports post-cutoff simulation results, and dis-
45 cusses implications for prospective evaluation.

46 **2. Literature Review**

47 Operations research has long addressed supply-chain uncertainty
48 through stochastic programming, robust optimization, and scenario-based
49 planning [4, 5, 6]. These methods formalize variability and recourse, but
50 operational inventory teams often face a more direct implementation prob-
51 lem: maintaining reorder points, safety stock, safety lead time, and order
52 quantities from incomplete historical evidence in ERP-driven planning en-
53 vironments [13, 14, 2, 3]. Static rules can be too coarse, while sophisticated
54 models can be difficult to explain and maintain. This motivates decision
55 support that keeps OR discipline while making material-level parameter
56 setting more transparent and auditable [7, 15].

57 Recent LLM research proposes natural-language interfaces, tool-using
58 agents, and pipelines that translate natural-language requests into opti-
59 mization model structures or executable code, including multi-agent systems
60 that make autonomous per-period inventory decisions [9, 16, 17, 18, 19, 20].
61 These approaches do not specify how inventory histories should be reduced
62 into demand regimes, lead-time diagnostics, service buffers, and feasible pol-
63 icy parameters with deterministic auditability.

64 InventOR follows this hybrid view. It supplies one filtered historical CSV
65 for the focal material plus an inline parameter block through an enterprise
66 LLM application. The intended prompt specification prohibits stochastic
67 simulation, machine-learning libraries, and unrestricted data fabrication,

but the stored run metadata do not independently identify the deployed prompt or runtime. Because LLM outputs can contain fabricated or unfaithful content [21], the executed downstream parser uses alias-aware lookups and nested-field resolution before requiring numeric positive reorder point and order quantity, numeric non-negative safety stock, and a resolvable simulator family. No project-specific model fitting or fine-tuning is performed. The design treats the LLM as an artifact-generation component inside a bounded decision process. Table 1 positions the framework relative to recent AI-supported OR and inventory-control work.

Table 1: Literature positioning: InventOR relative to recent AI-supported OR contributions for inventory and supply chain decision support.

Study	LLM role	Policy class	Demand model	Analytical result	Evaluation
Huang et al. [16]	NL→code	General OR	Not fied	speci- None	Benchmark
Ahmaditeshnizi et al. [18]	NL→model	General OR	Not fied	speci- Solver-linked	Benchmark
Zhang et al. [19]	Reasoning agent	General OR	Not fied	speci- Model solve	and Case studies
Quan and Liu [20]	Multi-agent control	Inventory	Sequential demand	Per-period de- cisions	Simulation
Prak et al. [22]	None	Inventory control	Compound Poisson / intermittent	Robust esti- mation	Empirical study
Babai et al. [23]	None	Forecasting-linked inventory	Regime-specific	Review thesis	syn- Review
Wasserkrug et al. [9]	Interface	General	Not fied	speci- None	Conceptual
Cohen et al. [17]	Vision	General SCM	Not fied	speci- None	Vision statement
InventOR	Prototype	(r, Q) executed; (s, S) im- plemented, unexer- cised	Artifact- defined values	Descriptive	Post-cutoff simulation

The table highlights the paper’s narrower contribution: material-level parameter artifacts evaluated by a deterministic policy-family simulator. Unlike InvAgent’s sequential multi-agent decisions, InventOR emits one sta-

tionary policy per material from a bounded historical package; unlike Opti-
MUS and OR-LLM-Agent, it does not formulate or solve general optimization models.

3. Methodology

The unit of analysis is a plant-material pair. The framework records a chain of custody from historical records to planner-facing policy values by separating deterministic input preparation, configured LLM inference, schema parsing, deterministic simulation, and human review [7, 24]. Each LLM session receives one filtered, material-scoped historical CSV as a session resource and a small inline parameter block containing current cost, MOQ, service, SAP reference, and material-level storage-capacity fields. No project-specific model weights are trained, fine-tuned, or updated. Because the stored metadata omit prompt hash, application version, and model/runtime identifier, this study does not claim that the exact deployed instruction configuration is independently reproducible. Supplementary File S1 reproduces the intended application-side system and tool prompt texts verbatim, together with their disclosed discrepancies against the executable simulator.

3.1. Problem Definition: Single-Item Continuous-Review Control with Stress-Adjusted Service Buffering

The executed task is a single-item, single-echelon continuous-review simulation with lost sales [13, 14, 25]. Each plant-material pair is treated independently; multi-item coupling, shared capacity, substitution, and production scheduling are outside the pilot. For the SAP-SLT-informed OR

comparator, pre-cutoff working-day demand yields mean demand $\bar{d}_i$, annualized demand $\bar{D}_i^{\text{ann}} = 250\bar{d}_i$, lead-time-demand dispersion, and stationary (R_i, Q_i) parameters. This comparator is not independent of ERP inputs because source SAP safety lead time enters effective lead time. For the SAP-derived arm, source safety stock is retained while R_i and Q_i are recomputed by the same executable routines. For the LLM arm, scoreable policy parameters originate directly in the artifact and pass alias-aware compatibility parsing plus strict executable-contract validation; the reference equations below do not reconstruct them. The simulator executes validated (r, Q) artifacts as fixed-quantity policies and validated (s, S) artifacts as order-up-to policies. In this study all 365 scoreable artifacts resolved to the (r, Q) family after validation, so the order-up-to path is an available but unexercised capability rather than an evaluated result.

Reference policy calculation. The following equations describe the executed SAP-SLT-informed OR comparator; they do not reconstruct direct LLM-emitted policies. The order quantity $Q_i > 0$ is rounded upward to an integer, floored at source MOQ, and then projected to its material-level storage limit when possible. The reorder point $R_i \geq 0$ triggers replenishment. The EOQ anchor minimizes the standard annual cost approximation

$$\min_{Q_i} TC_i(Q_i; SS_i) = h_i \left(\frac{Q_i}{2} + SS_i \right) + K_i \frac{\bar{D}_i^{\text{ann}}}{Q_i},$$

where SS_i is arm-specific safety stock, $h_i = \text{holding_cost_rate}_i \cdot \text{unit_cost}_i$, and $K_i = \text{ordering_cost}_i$. The implementation first uses $Q_i = \max\{\text{MOQ}_i, \lceil \sqrt{2K_i(250\bar{d}_i)/h_i} \rceil\}$ when demand and ordering cost are positive, and MOQ otherwise. For every arm, a supplied capacity requires $\text{MOQ}_i \leq Q_i$ and $SS_i + Q_i \leq \text{max_storage_units}_i$; comparator quantities

above the residual limit are reduced, while pairs with $SS_i + \text{MOQ}_i$ above the limit are excluded from the common cohort. The SAP-SLT-informed OR safety stock is $SS_i = \lceil z_{\alpha_i} \sigma_{LTD,i} \rceil$, with $R_i = \lceil \bar{d}_i L_i^{\text{eff}} + SS_i \rceil$. The SAP-derived arm substitutes source SAP safety stock into the same reorder-point expression. No lot-size multiple or stress coefficient is executed. The primary holding/order cost has a zero shortage penalty; positive lost-sales penalties are sensitivity analyses.

Core constraints. Inventory evolves under a lost-sales continuous-review rule on the working-day decision set $\mathcal{W}$. All arms share the first observed post-cutoff on-hand inventory; the SAP-SLT-informed OR $R_i + Q_i$ is used only when that observation is unavailable. The initial on-order pipeline is empty. Duplicate same-material, same-date rows are aggregated by summing demand, receipts, corrections, and scrap and averaging positive lead times, but the simulator consumes only demand and excludes observed receipts, corrections, and scrap. The simulator rounds mean source lead time and schedules every modeled order by that number of subsequent working observations:

$$I_{i,t} = \max(0, I_{i,t-1} + a_{i,t} - d_{i,t}), \quad t \in \mathcal{W},$$

where $a_{i,t}$ is the quantity whose working-day due observation is no later than t . Scheduled arrivals are added first, realized demand is served subject to lost sales, and the post-demand inventory position is computed as on-hand plus outstanding orders. For a validated (r, Q) artifact, a fixed quantity is ordered when

$$\delta_{i,t} = \mathbf{1}\{\text{IP}_{i,t}^+ \leq R_i\}, \quad \text{IP}_{i,t}^+ = I_{i,t} + \sum_{s: \text{due}(s) > t} o_{i,s},$$

150 with $o_{i,t} = Q_i \delta_{i,t}$. For a validated (s, S) artifact, the same threshold test
 151 uses s_i , and an order of $S_i - \text{IP}_{i,t}^+$ is placed when positive; no artifact in
 152 this study triggered that branch. Orders due after the observed horizon
 153 remain in transit rather than being forced onto the final observation. Pa-
 154 rameters remain stationary. Shared observed initialization and the empty
 155 initial pipeline are part of the executed comparison design.

156 The reorder point targets a nominal cycle service level $\alpha_i \in [0.5, 0.999]$
 157 after numeric clamping. Under the normal approximation,

$$\mathbb{P}\left(D_{i,L_i^{\text{eff}}} \leq R_i\right) \geq \alpha_i \iff R_i \geq \bar{d}_i L_i^{\text{eff}} + z_{\alpha_i} \sigma_{LTD,i}.$$

158 The reported backtest metric is the ex-post fill rate, which is distinct from
 159 the nominal cycle-service target because it also depends on Q_i and realized
 160 demand [26].

161 Lead-time-demand variance is propagated under the demand and lead-
 162 time independence approximation,

$$\sigma_{LTD,i}^2 = L_i^{\text{eff}} \sigma_{D,i}^2 + \bar{d}_i^2 \sigma_{L,i}^2.$$

163 The executed SAP-SLT-informed OR buffer is

$$SS_i^{\text{OR}} = z_{\alpha_i} \sigma_{LTD,i},$$

164 with values rounded upward:

$$SS_i^{\text{OR}} = \lceil z_{\alpha_i} \sigma_{LTD,i} \rceil, \quad R_i = \left\lceil \bar{d}_i L_i^{\text{eff}} + SS_i^{\text{OR}} \right\rceil.$$

165 The executable statistics add non-negative source SAP safety-lead-time days

166 to mean lead time when constructing comparator lead-time-demand disper-
167 sion (Section 3.5 discusses this confound). LLM-emitted values remain ar-
168 tifact values accepted by alias-aware parsing and strict executable-contract
169 validation.

170 *Parser and review boundary.* Artifact policy labels determine simulator
171 routing after validation. The parser resolves compatibility aliases, then
172 requires a valid (r, Q) tuple or explicit s and S values for an (s, S) label;
173 it rejects malformed, infeasible, or internally inconsistent artifacts. It does
174 not execute label-specific optimization or deterministic escalation. Because
175 no artifact supplied explicit and internally consistent s and S values, every
176 scored policy in this study was executed as (r, Q) ; the reported evidence
177 therefore concerns fixed-quantity control only. Planner review remains
178 required before any operational use.

179 3.2. Historical Input Schema and Canonical Analysis Table

180 The source table is grouped by plant and material, but the uploaded
181 material CSV contains only the subset listed in Table 2. Plant and material
182 identifiers are carried by the run key and inline block rather than repeated in
183 the uploaded rows. The inline block also passes current cost, MOQ, service,
184 SAP reference, working-day demand, lead-time, and material-level storage-
185 capacity variables. Deterministic preprocessing supplies input normaliza-
186 tion, eligibility filters, and simulator statistics. Stock-flow fields omitted
187 from the uploaded subset are not analyzed by the executed LLM path.

Table 2: Fields included in the material-scoped CSV uploaded by the executed runner.

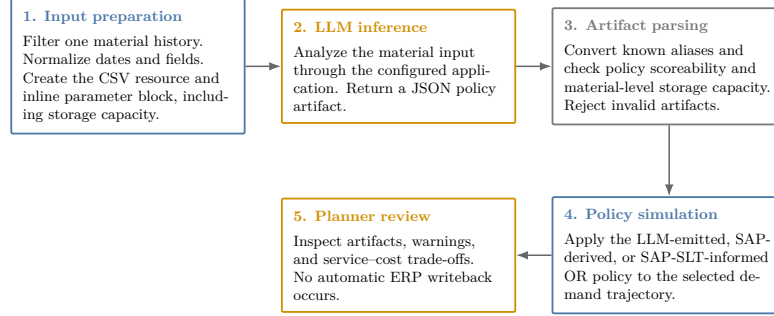
Column	Type	Description and method use
date	date / datetime	Calendar observation date; time ordering, aggregation, and diagnostics
actual_demand	numeric	Realized demand; mean, variability, intermittency, trend, and regime classification
delivery_time	numeric	Recorded replenishment lead time; rounded and scheduled as subsequent working observations by the simulator
minimum_order_quantity	numeric	Minimum replenishment quantity; order-size feasibility constraint
safety_stock_units	numeric	Existing safety stock; benchmarking against computed recommendations
safety_time_ workdays_actual	numeric	Source SAP safety-lead-time field; currently enters comparator effective lead time
supplier_plant_ distance	numeric	Supplier-to-plant distance; contextual lead-time-risk information
reliability_calc	numeric	Source supplier or lane reliability context
transportation_network	string	Source transportation-network context
service_level_target	numeric	Source service target used by comparator calculations
unit_cost	numeric	Purchase or standard cost per unit in the source dataset’s internal cost denomination; inventory-value proxy and input to holding-cost derivation and EOQ calculation
holding_cost_rate	numeric	Annual carrying-cost rate as a decimal (e.g., 0.20 for 20% of unit value); used to derive the annual cost of carrying one unit in stock, balance replenishment frequency against stock exposure, and compute economic order quantity
ordering_cost	numeric	Fixed replenishment cost in the source dataset’s internal cost denomination; input to EOQ calculation and simulated material-horizon holding/order cost
max_storage_units	numeric	Material-level upper bound applied to all evaluated arms: $MOQ \leq Q$ and $SS + Q \leq \text{max_storage_units}$
is_working_day	binary	Working-day flag; working-day demand statistics and non-zero demand share

3.3. Analytical Workflow

The workflow transforms a historical material snapshot into a policy artifact through deterministic preparation, LLM inference, parsing, and simulation (Figure 1).

InventOR prototype workflow

Executed material-level path from prepared history to planner-reviewable simulation output.



Deterministic components reproduce simulation results from stored artifacts; LLM inference itself may vary.

Figure 1: Prototype workflow from material input preparation to policy artifact and simulation output.

192 The runner uploads a normalized material CSV and inline block; the
 193 configured application returns a JSON artifact. The parser converts ac-
 194 cepted fields to an executable (r, Q) or (s, S) policy and rejects malformed
 195 or infeasible outputs. Missing or invalid artifacts are recorded and retried.
 196 The simulator evaluates accepted LLM values, SAP-derived values, and the
 197 SAP-SLT-informed OR policy on the common horizon. An aggregate-only
 198 manifest records active-input, evaluation-code, and stored-artifact hashes;
 199 it explicitly records unavailable historical deployed prompt, application,
 200 model/runtime, setting, and timestamp metadata. Current runner code
 201 records these fields for future runs when the corresponding environment vari-
 202 ables are supplied. The author declares retrospectively that the evaluated
 203 artifacts were produced with an Anthropic Claude Sonnet 4.6 model behind
 204 the enterprise application, using a code-interpreter style Python execution
 205 tool; this declaration is recorded in the manifest as unverified because no
 206 deployed model identifier or generation settings were persisted at run time,
 207 and it must not be read as machine-verified provenance. Stored inputs,

artifacts, and code reproduce a recorded simulation, but not a fresh LLM completion; planners retain implementation authority.

3.4. *Artifact Completeness and Future Readiness*

The completeness audit asks whether each eligible pair has a scoreable artifact after retries; it does not estimate first-pass reliability, explanation completeness, or deployment readiness. The empirical section is a retrospective audit of an already-executed workflow rather than a controlled experiment: the generation cohort was produced before the evaluation protocol was finalized, the deployed configuration was not frozen or instrumented in advance, and no treatment was assigned. The post-cutoff simulation uses artifacts generated exclusively from pre-cutoff inputs and is descriptive rather than a controlled prospective trial [27, 28]. A future evaluation must freeze the deployed configuration, model/runtime identifiers, access controls, and approval workflow.

3.5. *Safety Lead Time Computation*

Safety lead time (SLT) is an ancillary ERP-facing design field. The prototype specification proposes combining transport, supplier, and quality lead-time variability with a reliability-deficit premium [13, 3], but the present release does not document estimators for those components. In the executed comparator code, non-negative source SAP safety-lead-time days are added to mean lead time before computing both SAP-derived and OR-computed dispersion and reorder points (this coupling is also flagged in Section 3.1). The OR arm is therefore labelled SAP-SLT-informed, and no separate SLT effect is identified.

232 3.6. *Post-Cutoff Policy Simulation*

233 The confidential operational extract contains 233,078 parsed CSV
234 records, 411 plant-material pairs, and three anonymized plants (Plant
235 A, Plant B, and Plant C). The parameter cutoff is 2026-03-10 and the
236 post-cutoff simulation horizon ends on 2026-08-27. Eligibility filters retain
237 365 plant-material pairs with sufficient pre-cutoff and post-cutoff evidence:
238 234 from Plant A, 65 from Plant B, and 66 from Plant C. Plant C has
239 no positive source working-day flags, so its calendar flags are derived
240 from same-date Plant A and Plant B votes; ties classify as working
241 days. This deterministic repair permits a common decision calendar;
242 because no alternative Plant C calendar exists, robustness is examined by
243 reporting calendar-day arrivals and by re-running the full comparison on
244 the native-calendar subset that excludes Plant C entirely. SAP-derived and
245 SAP-SLT-informed OR parameters use pre-cutoff information. Stored LLM
246 artifacts were generated from cutoff-bounded pre-cutoff inputs, so every
247 arm is evaluated on the same post-cutoff horizon. Nineteen pairs cannot
248 meet comparator safety stock plus MOQ within their supplied capacity;
249 these are excluded before the common 346-pair comparison.

250 The simulated arms are: (i) SAP-derived, retaining source SAP safety
251 stock while recomputing reorder point and EOQ order quantity from pre-
252 cutoff statistics; (ii) SAP-SLT-informed OR, using the same statistics, source
253 safety-lead-time input, and feasibility rules but without LLM inference; and
254 (iii) LLM-emitted, a validated (r, Q) or (s, S) artifact. The simulator is
255 continuous-review and lost-sales: all arms begin with the same first observed
256 post-cutoff on-hand inventory, using the OR $ROP + Q$ only as a shared fall-
257 back. The initial on-order pipeline is empty. On each working row, modeled
258 arrivals due after rounded mean working-day lead time are received, demand

is served subject to lost sales, post-demand inventory position is checked, and a policy-specific order is placed if its threshold is met. Observed post-cutoff receipts, corrections, and scrap are excluded. This common startup design may create cutoff transients. Metrics are fill rate, aggregate demand-weighted fill rate, materials above 95% fill, zero-stockout materials, stockout days, average on-hand inventory, and simulated material-horizon cost.

For policy p and material i , the cost summaries use the simulated horizon of T_i working days. Let $\bar{I}_{i,p}$ be average on-hand inventory, h_i annual holding cost per unit, K_i fixed ordering cost, and $n_{i,p}^{\text{orders}}$ simulated orders. The implementation uses a constant 250 annual working days and computes

$$TC_{i,p} = h_i \bar{I}_{i,p} \frac{T_i}{250} + K_i n_{i,p}^{\text{orders}}.$$

This is a simulated material-horizon holding/order cost in the source dataset’s internal cost denomination, not an audited annual financial or booked accounting result. Table 3 reports arithmetic means across the cohort with zero shortage penalty. Sensitivities add lost-sales penalties of one and five times source unit cost; cost values must be read jointly with fill rate and stockout days.

4. Results

4.1. Completion and Artifact Validity

The final execution produced parser-scoreable artifacts for all 365 eligible plant-material pairs with zero missing and zero parser-rejected outputs after retry handling. Every artifact passes alias-aware parsing, nested-field resolution, and capacity validation against the supplied material-level storage bound; all resolve to the (r, Q) family after validation.

4.2. Main Three-Arm Exploratory Simulation

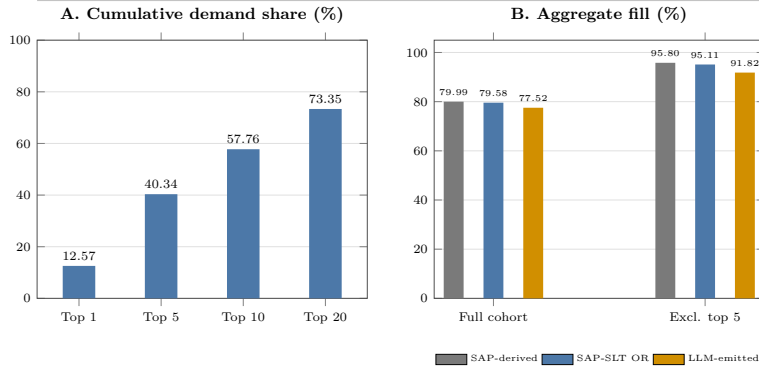
Table 3 reports three policy arms on the same 346 common-feasible material trajectories under working-day post-cutoff simulation. All arms satisfy identical MOQ and supplied per-material storage rules; 19 otherwise eligible pairs whose comparator safety stock plus MOQ exceed capacity are excluded.

Table 3: Post-cutoff working-day simulation on 346 common-feasible plant-material pairs. All arms use pre-cutoff parameterization or cutoff-bounded inputs and satisfy identical MOQ and supplied storage rules. Cost is mean simulated material-horizon holding/order cost with zero shortage penalty. SO days is the total count of stockout days across the cohort.

Policy	<i>n</i>	Mean FR (%)	Agg. FR (%)	$\geq 95\%$ (%)	Zero SO (%)	Mean cost	SO days
SAP-derived	346	98.38	79.99	94.51	80.35	3,184.20	741
SAP-SLT OR	346	98.27	79.58	93.93	88.73	3,852.22	799
LLM-emitted	346	97.94	77.52	92.49	81.50	2,848.96	936

Demand concentration and fill sensitivity

Largest-demand materials strongly influence full-cohort aggregate fill.



Generated from the working-days primary backtest; common-feasible cohort $n=346$.

Figure 2: Demand concentration and aggregate-fill sensitivity. Panel A reports cumulative demand share for the largest materials. Panel B compares full-cohort aggregate fill with the value after excluding the five largest-demand materials.

Relative to SAP-derived, LLM-emitted policies have 2.48 percentage points lower aggregate fill, 335.24 lower mean holding/order cost, and 195

290 additional stockout days. Relative to SAP-SLT-informed OR, they have 2.07
291 percentage points lower aggregate fill, 1,003.26 lower mean holding/order
292 cost, and 137 additional stockout days. These values describe one simulated
293 cohort, not a definitive ranking or an economic preference.

294 Aggregate evidence is demand-concentrated. The largest 1, 5, 10, and
295 20 materials contribute 12.57%, 40.34%, 57.76%, and 73.35% of post-cutoff
296 demand. Excluding the five largest-demand materials raises aggregate fill
297 to 91.82% for LLM-emitted, 95.80% for SAP-derived, and 95.11% for SAP-
298 SLT-informed OR, versus 77.52–79.99% in the full cohort (Figure 2). Thus,
299 the largest-demand items drive aggregate fill and should not be obscured by
300 cohort averages.

301 At material level, LLM-emitted fill is not lower than SAP-derived for
302 314 of 346 materials (90.8%), while its cost is no higher for 238 (68.8%);
303 both conditions hold for 211 (61.0%). Against SAP-SLT-informed OR,
304 the corresponding counts are 309 (89.3%), 303 (87.6%), and 269 (77.7%).
305 Paired LLM-minus-comparator fill differences have first quartile, median,
306 and third quartile of 0.00 percentage points for both comparators, reflecting
307 many exact fill ties. Corresponding cost differences are -259.24 , -47.17 ,
308 and 34.96 versus SAP-derived and -787.83 , -209.32 , and -25.26 versus
309 SAP-SLT-informed OR. A deterministic 10,000-resample paired bootstrap
310 gives LLM-minus-SAP fill and cost means of -0.44 percentage points (95%
311 interval $[-0.99, -0.02]$) and -335.24 (95% interval $[-629.16, -105.29]$);
312 against SAP-SLT-informed OR, the corresponding values are -0.33
313 percentage points ($[-0.89, 0.14]$) and $-1,003.26$ ($[-1,363.81, -705.03]$).
314 These descriptive resampling intervals and aggregate results jointly show
315 service–cost trade-offs rather than uniform policy ranking.

316 The artifact audit covers all 365 scoreable policies: normalized policy-

317 family agreement is 90.68%, and 42 artifacts meet a prespecified diagnostic-
 318 outlier rule (CV absolute error above 0.75 or LLM-to-pipeline safety-stock
 319 ratio outside $[0.5, 2.0]$). Removing the 23 flagged artifacts present in the
 320 common cohort raises LLM aggregate fill from 77.52% to 79.55%, while
 321 SAP-derived reaches 80.08% and SAP-SLT-informed OR reaches 79.70%;
 322 this post-hoc diagnostic subset does not establish a causal correction. Re-
 323 placing primary working-day receipt offsets with calendar-day offsets gives
 324 aggregate fills of 77.77%, 80.25%, and 79.84%, respectively. Removing the
 325 one plant whose source working-day flags are entirely absent, and whose
 326 calendar is therefore imputed, leaves 280 native-calendar pairs with aggre-
 327 gate fills of 74.03% for LLM-emitted, 76.55% for SAP-derived, and 76.35%
 328 for SAP-SLT-informed OR, mean material-level fills of 98.19%, 98.71%, and
 329 98.54%, zero-stockout shares of 81.79%, 88.21%, and 90.71%, and mean
 330 holding/order costs of 2,893.68, 3,471.66, and 4,168.76. In this native-
 331 calendar subset both comparators retain higher fill and higher zero-stockout
 332 shares than the LLM arm, so the imputed calendar does not manufacture the
 333 observed service gap. With a lost-sales penalty of one times unit cost, mean
 334 total costs are 1,688,922.75, 1,631,321.28, and 1,692,746.91, respectively; at
 335 five times unit cost, they are 8,433,217.92, 8,143,869.56, and 8,448,325.64.
 336 Thus, zero-penalty holding/order cost cannot establish economic superiority.
 337 These sensitivities do not resolve empty-pipeline, excluded-flow, or calendar-
 338 repair limitations.

339 5. Discussion

340 Three findings carry the principal scientific contribution. First, 100%
 341 of 365 eligible plant-material pairs yield scoreable, capacity-feasible policy

artifacts after retry handling a complete-cohort result that eliminates silent dropout and establishes a benchmark for future LLM-based inventory studies. Second, all scored artifacts resolve to the (r, Q) family; the LLM constructs every policy value from only the material-scoped history and parameter block, while both comparators are warm-started from ERP settings that embed accumulated planner judgment. The observed aggregate-fill deficit of roughly two percentage points, and the higher stockout-day count (936 versus 741 for SAP-derived and 799 for SAP-SLT-informed OR), are therefore cold-start characteristics of a generator that receives no safety-stock legacy or safety-lead-time input. An ERP-independent cold-start OR arm is the comparator required to isolate generator quality and is specified as a future-work requirement. Third, positive lost-sales penalty sensitivities remove the zero-penalty cost ranking, confirming that the apparent cost advantage is driven by the penalty assumption rather than by generator quality.

The evaluation protocol itself is the reusable contribution. An aggregate-only manifest hashes active inputs, evaluation code, and stored artifacts, while explicitly flagging unavailable provenance fields. The input-contract specification, alias-aware parser, capacity-validation rules, and the full exclusion log form a reproducible boundary within which any LLM or OR generator can be substituted and scored under identical rules. The tracked historical 315-material snapshots are byte-identical rather than distinct generation replicates; the two replication runs launched for this manuscript provide the cohort-scale generation-variance estimate that those snapshots cannot.

The study is bounded by its single anonymized three-plant dataset, one generation configuration, and a post-cutoff simulation that excludes observed receipts, corrections, and scrap with an empty initial pipeline. These

are explicit design choices, not undisclosed weaknesses. The contribution is the protocol and its first complete execution, not a performance claim.

6. Conclusion

This paper presents *InventOR*, an applied AI/OR prototype that treats LLMs as candidate-policy generators within deterministic preparation, parsing, and simulation. The framework preserves an inspectable link between material histories, policy fields, and simulated outcomes without project-specific model training or fine-tuning.

The executed artifact set contains 365 scoreable policies generated from cutoff-bounded, pre-cutoff material histories. Every accepted LLM policy satisfies its material-level storage bound. On the 346-pair cohort where all arms satisfy identical MOQ and supplied storage rules, LLM-emitted aggregate fill is 77.52%, mean material-level fill is 97.94%, mean simulated material-horizon holding/order cost is 2,848.96, and stockout days total 936 under working-day receipt scheduling. Positive shortage-cost sensitivities remove the apparent cost advantage. These values are descriptive simulated-comparison evidence, not repeated-inference or controlled prospective-trial evidence. The framework establishes a reviewable prototype and protocol for subsequent prospective evaluation, not a deployment or adoption claim.

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395 **Declaration of Competing Interest**

396 The author declares no known competing financial interests or personal
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398 paper.

399 **Data and Code Availability**

400 The raw operational data and material-level artifacts cannot be publicly
401 released because they contain commercially sensitive material, supplier, and
402 planning information. A confidentiality-screened release package contain-
403 ing analysis code, the anonymized variable schema, aggregate evaluation
404 results, manuscript sources, and aggregate integrity manifest is available
405 at github.com/DressPD/inventor_public. The release package is addi-
406 tionally deposited in a Zenodo archive whose version-specific DOI will be
407 registered at acceptance and cited in the final version of record; the GitHub
408 repository is the development mirror and the Zenodo deposit is the citable
409 archival copy. Supplementary File S1 accompanies this submission and con-
410 tains the intended application-side prompt specification. The historical de-
411 ployed application configuration, model/runtime identifier, inference dates,
412 and prompt hashes were not preserved and are not claimed as available; the
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420 remain the responsibility of the author.

421 Declaration of Generative AI and AI-Assisted Technologies in the 422 Writing Process

423 During manuscript preparation, the author used OpenAI GPT-5.6
424 through OpenCode for language editing, consistency checks, citation
425 review, and LaTeX proofing support; this is a writing-assistance tool and
426 is distinct from the Anthropic Claude Sonnet 4.6 deployment evaluated in
427 the research workflow. The author reviewed and edited the content after
428 use and takes full responsibility for the content of the published article.
429 Figures were rendered deterministically from author-reviewed LaTeX and
430 Python source; generative AI did not directly create or manipulate image
431 content. The research workflow evaluated in the paper is itself the object
432 of study and is described separately in the methodology.

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